

```
C:\WINDOWS\system32\cmd. x + v

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>notepad app\main.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>notepad tests\test_main.py

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>python -m pytest -q
...
===== warnings summary =====
.venv\Lib\site-packages\fastapi\testclient.py:1
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline\.venv\Lib\site-packages\fastapi\testclient
.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient' is deprecated; install 'httpx2' instead.
    from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
3 passed, 1 warning in 11.35s

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>ruff check .
All checks passed!

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>python -m uvicorn app.main:app --rel
oad
INFO: Will watch for changes in these directories: ['C:\\Users\\Sanjana CA\\OneDrive\\Documents\\CynarisInternship\\W12D4_End_to_
End_MLOps_Pipeline']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [4844] using StatReload
INFO: Started server process [23736]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: 127.0.0.1:56278 - "GET /docs HTTP/1.1" 200 OK
INFO: 127.0.0.1:56278 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 127.0.0.1:49958 - "POST /predict HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [23736]
INFO: Stopping reloader process [4844]
```

```
C:\WINDOWS\system32\cmd. x + v

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>docker build -t w12d4-mlops-api:late
st .
[+] Building 185.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 264B
=> [internal] load metadata for docker.io/library/python:3.12-slim
=> [internal] load .dockerignore
=> => transferring context: 111B
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:f77ac9e44ae96ef2c90b8053ea08c31f8be030f824196b0ae4db6d462c84e51f
=> => resolve docker.io/library/python:3.12-slim@sha256:f77ac9e44ae96ef2c90b8053ea08c31f8be030f824196b0ae4db6d462c84e51f
=> [internal] load build context
=> => transferring context: 3.05kB
=> CACHED [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY app ./app
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:ae9479934bb13e18c9318464a33e012dbc5a8516ba7393beaeade51536c6a10
=> => exporting config sha256:44573fbbcab101a72adc32ae248f151fe43f510075af6080633d52c9cce0a8ec
=> => exporting attestation manifest sha256:23be788750990b8b5a04991e4a54989af342ffef2b8ee759992772f700ee1823
=> => exporting manifest list sha256:6ab9f10b8e24b4b7ea83fdc87a9f25e815dbb5deb17ab46c7663658c293098bd
=> naming to docker.io/library/w12d4-mlops-api:latest
=> unpacking to docker.io/library/w12d4-mlops-api:latest

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>docker images
Info -> U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
mlops-ml-api:latest 068632f888ea 627MB 147MB U
production-rag-api:latest ca25408bcb55 310MB 75.8MB
w12d2-ml-api:latest 6bbb1d5f399d 604MB 140MB
w12d3-ml-monitoring:latest 76148c63f2a2 591MB 137MB U
w12d4-mlops-api:latest 6ab9f10b8e24 572MB 133MB

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>docker run -d --name w12d4-mlops-con
tainer -p 8000:8000 w12d4-mlops-api:latest
```

```
C:\WINDOWS\system32\cmd. x + v
password: ${ secrets.DOCKERHUB_TOKEN }}

- name: Build and push Docker image
  run: |
    docker build -t ${ secrets.DOCKERHUB_USERNAME }}/w12d4-mlops-api:latest .
    docker push ${ secrets.DOCKERHUB_USERNAME }}/w12d4-mlops-api:latest
(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>ruff check .
All checks passed!

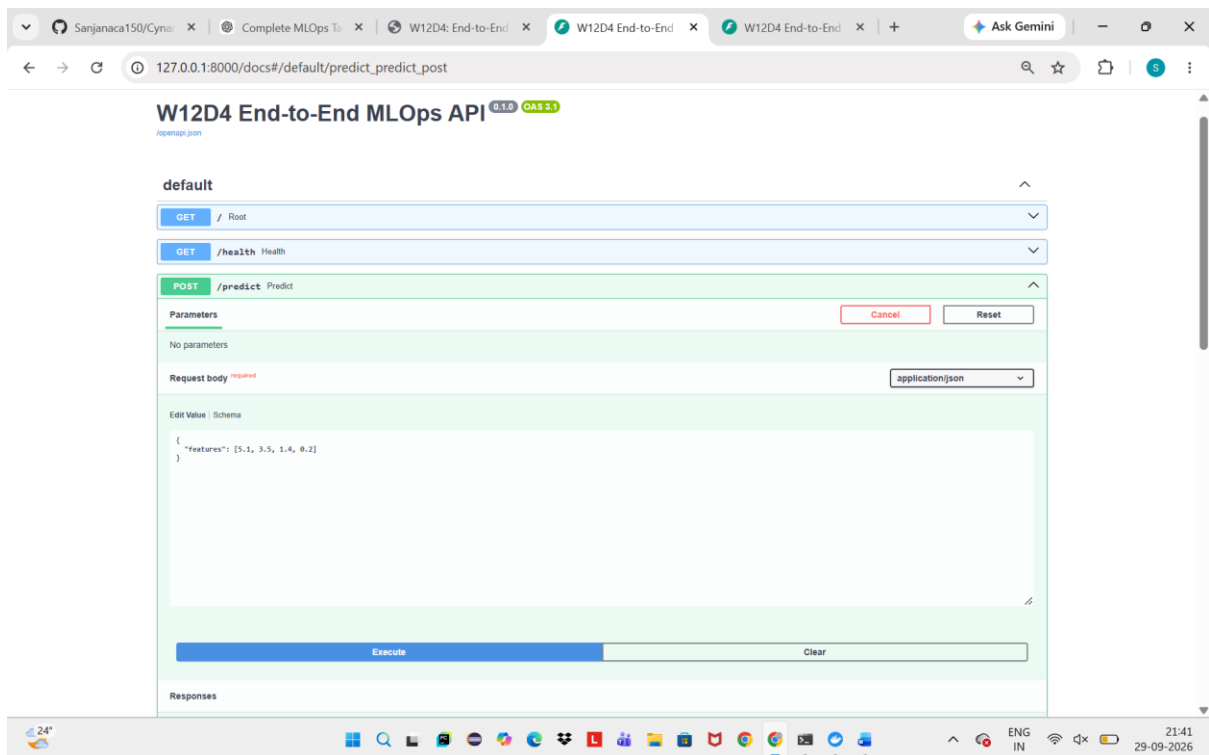
(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>pytest -q
[100%]
===== warnings summary =====
.venv\Lib\site-packages\fastapi\testclient.py:1
C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline\.venv\Lib\site-packages\fastapi\testclient
.py:1: StarletteDeprecationWarning: Using 'httpx' with 'starlette.testclient' is deprecated; install 'httpx2' instead.
  from starlette.testclient import TestClient as TestClient # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
3 passed, 1 warning in 8.65s

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
d314d495d4c5   w12d4-mlops-api:latest              "uvicorn app.main:ap..." 4 hours ago    Up 4 hours    0.0.0.0:8000->8000/tcp, [::]:8000->8000/t
cp   w12d4-mlops-container

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>docker logs w12d4-mlops-container
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: 172.17.0.1:45286 - "GET /docs HTTP/1.1" 200 OK
INFO: 172.17.0.1:45286 - "GET /openapi.json HTTP/1.1" 200 OK
INFO: 172.17.0.1:45278 - "GET /health HTTP/1.1" 200 OK
INFO: 172.17.0.1:56830 - "POST /predict HTTP/1.1" 200 OK

(.venv) C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W12D4_End_to_End_MLOps_Pipeline>
```



Sanjanaca150/Cynr Complete MLOps T W12D4: End-to-End W12D4 End-to-End W12D4 End-to-End Ask Gemini

127.0.0.1:8000/docs#/default/predict_predict_post

Responses

Curl

```
curl -X 'POST' \
  "http://127.0.0.1:8000/predict" \
  -H 'Accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "features": [5.1, 3.5, 1.4, 0.2]
  }'
```

Request URL

http://127.0.0.1:8000/predict

Server response

Code Details

200

Response body

```
{
  "prediction": 0,
  "class_name": "setosa"
}
```

Response headers

```
content-length: 38
content-type: application/json
date: Tue, 29 Sep 2026 11:41:17 GMT
server: waitress
```

Responses

Code	Description	Links
200	Successful Response	No links
	Media type: application/json	
	Content Accept header:	
	Example Value Schema	
	"string"	
422	Validation Error	No links
	Media type: application/json	
	Example Value Schema	

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Sanjanaca150/Cynr Complete MLOps T W12D4: End-to-End W12D4 End-to-End W12D4 End-to-End Ask Gemini

127.0.0.1:8000/docs#/default/predict_predict_post

Responses

Code	Description	Links
200	Successful Response	No links
	Media type: application/json	
	Content Accept header:	
	Example Value Schema	
	"string"	
422	Validation Error	No links
	Media type: application/json	
	Example Value Schema	

```
{
  "detail": {
    "loc": [
      "string",
    ],
    "msg": "string",
    "type": "string",
    "input": "string",
    "ctx": {}
  }
}
```

Schemas

HTTPValidationError > Expand all object

PredictionRequest > Expand all object

ValidationError > Expand all object

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Sanjanaca150/Cynr x Complete MLOps T W12D4: End-to-End x W12D4 End-to-End x W12D4 End-to-End x + Ask Gemini

localhost:8000/docs#/default/predict_predict_post

W12D4 End-to-End MLOps API 0.1.0 QAS 3.1

/openapi.json

default

GET / Root

GET /health Health

Parameters Cancel

No parameters

Execute **Clear**

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8000/health' \
  -H 'accept: */*'
```

Request URL

http://localhost:8000/health

Server response

Code	Details
200	

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Sanjanaca150/Cynr x Complete MLOps T W12D4: End-to-End x W12D4 End-to-End x W12D4 End-to-End x + Ask Gemini

localhost:8000/docs#/default/predict_predict_post

Server response

Code	Details						
200	Response body <pre>{ "status": "healthy", "model": "RandomForestClassifier" }</pre> Response headers <pre>content-length: 53 content-type: application/json date: Tue, 29 Sep 2026 12:13:22 GMT server: uvicorn</pre> Responses <table><thead><tr><th>Code</th><th>Description</th><th>Links</th></tr></thead><tbody><tr><td>200</td><td>Successful Response</td><td>No links</td></tr></tbody></table> Media type: application/json Controls Accept header. Example Value Schema <pre>"string"</pre>	Code	Description	Links	200	Successful Response	No links
Code	Description	Links					
200	Successful Response	No links					

POST /predict Predict

Parameters Cancel Reset

No parameters

Request body required application/json

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Sanjanaca150/Cynr... Complete MLOps T... W12D4: End-to-End x W12D4 End-to-End x W12D4 End-to-End x + Ask Gemini

localhost:8000/docs#/default/predict_predict_post

Request body **required** application/json

Edit Value | Schema

```
{
  "features": [5.1, 3.5, 1.4, 0.2]
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  http://localhost:8000/predict' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "features": [5.1, 3.5, 1.4, 0.2]
  }'
```

Request URL

```
http://localhost:8000/predict
```

Server response

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Sanjanaca150/Cynr... Complete MLOps T... W12D4: End-to-End x W12D4 End-to-End x W12D4 End-to-End x + Ask Gemini

localhost:8000/docs#/default/predict_predict_post

Server response

Code	Details
200	Response body <pre>{ "prediction": 0, "class_name": "setosa" }</pre> Response headers <pre>content-length: 38 content-type: application/json date: Tue, 29 Sep 2026 12:34:00 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links
	Media type: application/json	
	Controls Accept header:	
	Example Value Schema	
	<pre>"string"</pre>	
422	Validation Error	No links
	Media type: application/json	
	Example Value Schema	
	<pre>{ "detail": [{ "loc": ["string", 0], "msg": "string", }] }</pre>	

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A screenshot of a web browser window displaying a REST client interface. The address bar shows the URL 'localhost:8000/docs#/default/predict_predict_post'. The page content shows a '422 Validation Error' response. The error message is 'string', which is highlighted in a dark box. Below the error message, there is a 'Media type' dropdown menu set to 'application/json'. Further down, the error details are shown in a dark box, displaying a JSON object: { 'detail': [{ 'loc': ['string', 0], 'msg': 'string', 'type': 'string', 'input': 'string', 'ctx': {} }] }. The browser's taskbar at the bottom shows various application icons and the system clock indicating 21:45 on 29-09-2026.